



WHAT IS IN THIS FOLDER

1 Computer Science at Calvin

Information about the degrees in Computer Science, Computer Science Plus and Data Science, and about the cybersecurity certificate. Our BCS is accredited by the Computing Accreditation Commission of ABET.

2 Meet the faculty

The professors who teach in the Computer Science Department, their research, and how to reach out to any of them directly.

3 A map for your four years

A semester-by-semester course plan you can tick off, one sheet for each degree.

IF YOU HAVE NEVER WRITTEN CODE

No previous experience in programming is needed! CS 108, the first course in the major, starts from the beginning. Math, science, English and some coding in high school leave you well equipped.

TAKE THE NEXT STEP



VISIT US

Sit in on a class and see the labs.
calvin.edu/visit



APPLY

calvin.edu/admissions/apply



ASK US ANYTHING

Write to any professor on the faculty sheet.
computing.calvin.edu



Scan for the department site, current student projects, and the courses you would take in your first year.



COMPUTER SCIENCE

DEPARTMENT DISTINCTIVES

At the Computer Science Department you can major in three ways: the Bachelor of Computer Science (BCS), Computer Science Plus (BS in CS), or Data Science (BS in DS). There are also minors in computer science, data science and data analytics, and a new certificate in cybersecurity. Students work beside professors who research artificial intelligence and publish on how computing is taught. A liberal arts core means you graduate able to write and communicate in addition to coding, and every professor is a committed Christian who brings their faith to bear on teaching and research.

BACHELOR OF COMPUTER SCIENCE (BCS)

The BCS is ABET-accredited, meeting the same standards as programs at major research universities. Coursework spans algorithms, artificial intelligence, security, databases, networks, and software engineering. Graduates move into software development, networking, and systems work, or on to graduate study at places like Johns Hopkins, MSU, Michigan Tech, Purdue and the University of Sydney.

COMPUTER SCIENCE PLUS (BS IN CS)

A full computer science degree paired with a second field, for students who want to bring computing to the subject they care about: business, psychology, languages, or another field entirely. Our Strategic Partners Council regularly reviews the curriculum so it tracks the technologies employers actually use, and students may take paid internships and do research with faculty.

DATA SCIENCE (BS IN DS)

Calvin was among the first Christian institutions to offer data science. Students combine computer science, statistics, and mathematics with four to five courses in a second discipline such as biology, geography, business, economics, or political science, then finish with a capstone project on real data.

CYBERSECURITY CERTIFICATE NEW FOR FALL 2026

Eighteen credit hours in cryptography, network defense, and system administration, designed around the NSA's Cyber Defense knowledge units and taught by a professor with three decades in network security. BCS students earn the certificate inside their existing major with no added credit hours. The Bureau of Labor Statistics projects this field will grow 21 percent between 2025 and 2035.





Derek Schuurman

PROFESSOR • CHAIR • NH 296

Focus: Faith and technology, embedded systems and IoT, robotics and computer vision

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- PhD, McMaster University
- BAsC and MASc in Electrical Engineering, University of Waterloo
- Writes on the ethics and philosophy of technology



Ken Arnold

ASSOCIATE PROFESSOR • NH 298

Focus: Data science, artificial intelligence and Christian faith

Email: ken.arnold@calvin.edu

- PhD in Computer Science, Harvard University, 2020
- SM, MIT Media Lab; BS in Electrical and Computer Engineering, Cornell
- Language models for writing, and public understanding of what AI can and cannot do



Eric Araújo

ASSOCIATE PROFESSOR • NH 293

Focus: Agent-based modelling, complex networks, computational social science

Email: eric.araujo@calvin.edu

- PhD in Computer Science, Vrije Universiteit Amsterdam
- MS, Federal University of Minas Gerais; BS, Federal University of Viçosa, Brazil
- Studies how behaviour, health and misinformation spread



Rocky Chang

PROFESSOR • NH 299A

Focus: Network security, internet infrastructure, computational thinking

Email: rocky.chang@calvin.edu

- PhD in Computer and Systems Engineering, Rensselaer Polytechnic Institute
- More than 30 years teaching computer science
- Leads the cybersecurity certificate



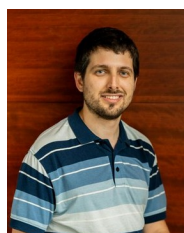
Victor Norman

ASSOCIATE PROFESSOR • NH 297

Focus: Computer science education, K-12 outreach, computing for Christian missions

Email: victor.norman@calvin.edu

- PhD and MS, Purdue University; BS, Calvin College
- 14 years in industry as a principal software engineer before teaching
- Works on applying the virtues to how we write programs



Fernando Santos

ASSISTANT PROFESSOR • NH 291

Focus: Data science, computational modelling, human-computer interaction

Email: fernando.santos@calvin.edu

- PhD in Computational Science, University of São Paulo
- Also works on the philosophy of technology
- Originally from Poços de Caldas, Brazil



Chris Wieringa

SYSTEMS ENGINEER • SB 319

Focus: The labs, the servers and the systems every course here runs on

Email: chris.wieringa@calvin.edu

- Runs the labs, servers and systems behind every computer science course

ALSO TEACHING WITH US

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COMPUTER SCIENCE DEPARTMENT

CHRIST-CENTERED UNDERSTANDING

Technology is not neutral. This is part of how we teach you to build new tools for the world. You will be challenged by the many discussions on ethics and philosophy of technology as you progress in your knowledge of computer science. Faith is not an appendix, but the core of what we learn together here.

PROFESSORS WITH OPEN DOORS

Every course in the major is taught by a professor, and their offices sit along one corridor in North Hall. Their doctorates come from Harvard, Purdue, McMaster, Rensselaer, Amsterdam and São Paulo, and several worked in industry first.

FOUR-YEAR PATH TO GRADUATION

The Bachelor of Computer Science is 124 credit hours when paired with the Calvin Core, so you graduate with a breadth of knowledge as well as depth in your field. The degree is ABET accredited, the standard employers and graduate schools look for.

ACADEMIC OPTIONS

DEGREES

Bachelor of Computer Science (BCS)

ABET accredited.

BS in Computer Science Plus

Computer science with an applied field of your choice.

BS in Data Science

Computer science and statistics, paired with a subject area.

MINORS

Data Science

Data Analytics

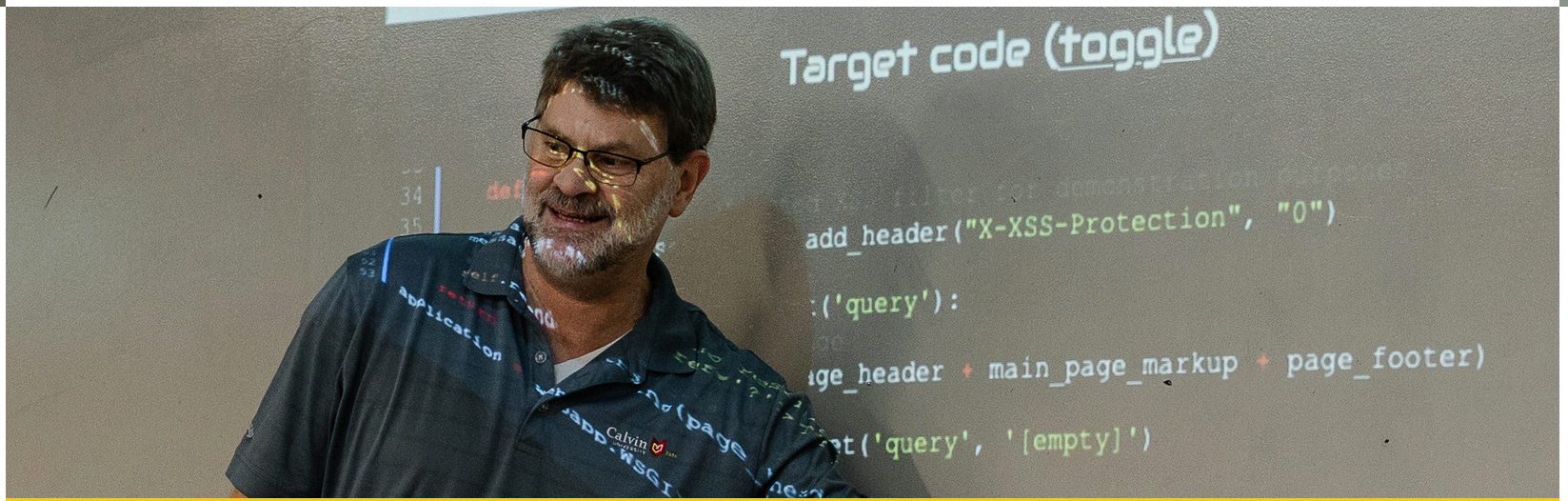
CERTIFICATE

Cybersecurity

Six courses, 18 credit hours, taken with any of the degrees.

PAIR WITH

Biology · Business · Economics · Geography · Political science · Psychology · Languages



WHAT YOU DO HERE

INTERNSHIPS THAT LEAD SOMEWHERE

Students work a paid internship at a local firm or a national one. For some students, internships become full-time jobs. Recent placements include AMD, Amazon Web Services, General Motors, Corewell Health, Navy Federal Credit Union, Meta, Vervint, Bindplane, EduStaff, Sonset Solutions, Wycliffe Bible Translators, and the Christian Classics Ethereal Library.



UNDERGRADUATE RESEARCH OPPORTUNITIES

You can join a professor's research already as an undergraduate student. Current research spans language models and everyday creativity, internet infrastructure and security, agent-based models of how misinformation and health behaviour spread, education tools, high performance computing, artificial intelligence, and embedded systems.



LIBERAL ARTS AND THE CALVIN CORE

Employers want people who can size up a problem, think it through, explain it in writing, and work well with colleagues, clients, and the people they report to. That is what the core builds, and it is woven throughout your undergraduate experience.



WHERE OUR GRADUATES GO

Alumni are at Google, Amazon, Meta, Steelcase, Corewell Health, Bank of America, the Washington Post, Amway and Pray.com. Others have gone on to graduate school at Johns Hopkins, MSU, Michigan Tech, Purdue, the University of Waterloo, and the University of Sydney, and some now teach at Grand Valley State University and George Mason University. Some have stayed and joined the Christian Classics Ethereal Library and hymnary.org.



EDUCATION & COURSEWORK

FOUNDATIONS: YEARS 1 AND 2

You start writing programs in your first semester, in CS 108, and CS 112 takes you into data structures the semester after. The first two years build the craft and the theory underneath it: programming, the machine below the language, and the mathematics that lets you prove an algorithm works rather than hope it does. Core courses run from the first year too, so the questions about what computing is for start early.



DEPTH: YEARS 3 AND 4

In the upper years you choose the direction: systems and networks, artificial intelligence and data, security, software engineering, or the applied area of your choice. The fourth year goes into one substantial project, the one employers and graduate schools ask about.



CALVIN HACKATHON

- Students form teams and create a project from scratch in 24 hours
- Students also compete off campus. A Calvin pair took first place at the Atomic Object games

CYBERSECURITY CERTIFICATE

- Six courses, 18 credit hours, taken with any degree
- Built around the Cyber Defense knowledge units published by the NSA and NCAE-C

SEMINARS

- People working in the field come in to talk about jobs, careers, and where computer science, data science and AI are heading
- They run through all four years, not only the final one

SENIOR PROJECT

- A full-year project for real
- Teams choose a project, then plan, research, build, test and deploy it over the year
- Some teams do a research study instead
- Every project ships a public website, and this year a poster too

The full course list for each degree is on the model program sheets in this folder, one per degree.

MORE THAN THE CODE

You write code in your first semester and build systems soon after, always thinking of the impact of your work in the light of a Christian Reformed tradition. Artificial intelligence has not retired that question, it has sharpened it. Faith holds together the craft of building something that works and the wisdom to know how to shape it.

ADVICE FOR HIGH SCHOOL

Take mathematics as far as your school offers it, and take English and the sciences seriously. This degree asks you to reason, write, and argue as much as to code. If your school teaches programming, that can provide additional preparation.

WHAT THE DEGREE LEADS TO

- Software engineering
- Artificial intelligence and machine learning
- Cybersecurity and computer forensics
- Data science and analytics
- Cloud architecture and infrastructure
- Databases and information systems
- Embedded systems

STUDENT ORGANIZATIONS

- Abstraction
- Women in Computing
- Google Developer Student Club

Game nights, Lego events, scavenger hunts and sticky-note pixel art.

MENTORED FROM THE START

Every student has a faculty mentor from the first semester. You meet through your four years, so there is someone who knows your work when you need advice or a reference.

CONFERENCES AND COMPETITIONS

Women in Computing takes students to the Grace Hopper Celebration, the largest gathering of women in computing anywhere, and faculty travel with them. Other conferences include:

- Calvin Hackathon
- GrrCon and GR Tech Week
- CCSC and the Create Conference



Bachelor of Computer Science

Degree accredited by the Computing Accreditation Commission of ABET.

FRESHMAN

FALL	14 hrs	SPRING	15 hrs
4 CS 108 Introduction to Computing F,S		4 CS 112 Introduction to Data Structures F,S	
4 MATH 161 Calculus: Modeling and Computation F,S		2 MATH 251 Discrete Mathematics I S	
0 CS 195 Introductory Computing Seminar F,S		0 CS 195 Introductory Computing Seminar F,S	
2 CORE 100 Community and Commitments		4 ENGL 101 Foundational Writing	
4 World Languages 1 Waived with two years of high school language at B or better		4 World Languages 2	
		1 Health and Movement 1	

SOPHOMORE

FALL	16 hrs	SPRING	16 hrs
4 CS 212 Data Structures and Algorithms F		4 CS 214 Programming Language Concepts S	
4 MATH 255 Introductory Linear Algebra F		2 MATH 252 Discrete Mathematics II S	
4 DATA 202 Introduction to Machine Learning F		4 STAT 243 Statistics S	
0 CS 195 Introductory Computing Seminar F,S		4 COMM 101 Oral Rhetoric	
4 Christian Foundations 1		2 Christian Foundations 2	

JUNIOR

FALL	16 hrs	SPRING	15 hrs
4 CS 262 Software Engineering F needs junior standing		4 CS 232 Operating Systems and Networking S	
4 ENGR 220 Introduction to Computer Architecture F		4 CS elective From the advanced list	
0 CS 295 Computing Seminar F,S		0 CS 295 Computing Seminar F,S	
4 Core course Knowledge and Understanding		1 Health and Movement 2	
4 Elective		4 Core course Knowledge and Understanding	
		2 Elective	

SENIOR

FALL	16 hrs	SPRING	16 hrs
2 CS 396 Senior Project in Computing F		2 CS 398 Senior Project in Computing II S	
0 CS 295 Computing Seminar F,S		4 CS 384 Perspectives on Computing S	
2 CS elective		4 CS elective	
4 Science elective Laboratory science		2 Science elective	
4 Core course Knowledge and Understanding		4 Elective	
4 Elective			

F offered in the autumn S offered in the spring · a full-time semester is about 16 hours
This is one way through the degree. Your adviser will build yours with you.

TO GRADUATE
124 hours

Computer Science Plus

Applied computer science in the field of your choice.

FRESHMAN

FALL	14 hrs	SPRING	15 hrs
4 CS 108 Introduction to Computing F,S		4 CS 112 Introduction to Data Structures F,S	
4 MATH 161 Calculus: Modeling and Computation F,S		2 MATH 251 Discrete Mathematics I S	
0 CS 195 Introductory Computing Seminar F,S		0 CS 195 Introductory Computing Seminar F,S	
2 CORE 100 Community and Commitments		4 ENGL 101 Foundational Writing	
4 World Languages 1 Waived with two years of high school language at B or better		4 World Languages 2	
		1 Health and Movement 1	

SOPHOMORE

FALL	16 hrs	SPRING	16 hrs
4 CS 212 Data Structures and Algorithms F		4 CS 214 Programming Language Concepts S	
4 DATA 202 Introduction to Machine Learning F		2 MATH 252 Discrete Mathematics II S	
4 Your applied field Twelve hours in total		4 STAT 243 Statistics S	
0 CS 195 Introductory Computing Seminar F,S		4 Core course Arts and Rhetoric	
4 Christian Foundations 1		2 Christian Foundations 2	

JUNIOR

FALL	16 hrs	SPRING	15 hrs
4 CS 262 Software Engineering F needs junior standing		4 CS 232 Operating Systems and Networking S	
4 ENGR 220 Introduction to Computer Architecture F		4 CS elective Six hours in total	
4 Your applied field		0 CS 295 Computing Seminar F,S	
0 CS 295 Computing Seminar F,S		1 Health and Movement 2	
4 Core course Knowledge and Understanding		4 Core course Knowledge and Understanding	
		2 Elective	

SENIOR

FALL	16 hrs	SPRING	16 hrs
2 CS 396 Senior Project, or CS 383 Practicum F		2 CS 398 Senior Project in Computing II S	
0 CS 295 Computing Seminar F,S		4 CS 384 Perspectives on Computing S	
4 Your applied field		2 CS elective	
4 Core course Knowledge and Understanding		4 Elective	
4 Elective		4 Elective	
2 Elective			

F offered in the autumn S offered in the spring · a full-time semester is about 16 hours
This is one way through the degree. Your adviser will build yours with you.

TO GRADUATE
124 hours

Computer science, mathematics and statistics together, to equip new professionals of this demanding field of work.

Data Science

FRESHMAN

FALL	14 hrs	SPRING	16 hrs
4 CS 108 Introduction to Computing F,S		4 CS 112 Introduction to Data Structures F,S	
4 MATH 161 Calculus F,S		4 STAT 243 Statistics S	
2 CORE 100 Community and Commitments		4 ENGL 101 Foundational Writing	
4 World Languages 1 Waived with two years of high school language at B or better		4 World Languages 2	

SOPHOMORE

FALL	16 hrs	SPRING	15 hrs
4 DATA 202 Introduction to Machine Learning F		4 STAT 341 Computational Bayesian Statistics S	
4 MATH 255 Introductory Linear Algebra F		4 Core course Arts and Rhetoric	
4 Core course Knowledge and Understanding		1 Health and Movement 1	
4 Christian Foundations 1		2 Christian Foundations 2	
		4 Elective	

JUNIOR

FALL	16 hrs	SPRING	15 hrs
4 STAT 245 Applied Data Analysis F		2 CS 375 Machine Learning S	
4 CS elective CS 212 recommended		2 CS 376 Advanced Machine Learning S	
4 Business elective Six hours in total		2 DATA 304 Visualizing Data and Models S	
4 Elective		2 DATA 385 Topics in Data Science S	
		1 Health and Movement 2	
		4 Core course Knowledge and Understanding	
		2 Elective	

SENIOR

FALL	16 hrs	SPRING	16 hrs
2 DATA 396 Senior Project in Data Science F		2 DATA 398 Senior Project in Data Science II S	
2 DATA 385 Topics in Data Science, second topic F		4 CS 384 Perspectives on Computing S	
4 Ethics elective PHIL 205 or BUS 392		2 Business elective	
4 Core course Knowledge and Understanding		4 Elective	
4 Elective		4 Elective	

F offered in the autumn S offered in the spring · a full-time semester is about 16 hours
This is one way through the degree. Your adviser will build yours with you.

TO GRADUATE
124 hours

